EC571: Digital VLSI Circuit Design

Homework 4 — Fall 2026

Total: 100 points
Due: October 30, 2026 at 11:59 pm ET

General Instructions

Use this worksheet to complete the homework. Please print this worksheet, write your solutions on the worksheet, take a picture of the worksheet, and submit it through Gradescope. If you prefer, you can download the Homework 4 worksheet on your tablet, complete the worksheet on your tablet, and submit the completed worksheet through Gradescope.

Show intermediate reasoning, calculations, and clearly labeled transistor-level schematics wherever requested. State the assumptions and models used. Answers without supporting analysis will receive limited credit. Unless otherwise stated, use the transistor and logic models discussed in Lectures 9 and 10. Do not assume an unstated circuit diagram: your diagrams should make all required connections explicit.

For Question 4, submit the specified Cadence Virtuoso screenshots, transient plots, tables, and transistor fingering-sweep graphs. Use AI tools only for concept review, not to generate final solutions or submission materials; include a one-sentence AI-use disclosure.

Question 1: Pseudo-NMOS Design Under Competing Constraints [20 points]

Consider implementing

$$Y = \overline{AB + C}$$

using pseudo-NMOS logic. The circuit consists of one always-on PMOS pull-up transistor and an NMOS pull-down network implementing the appropriate logic function.

Assume:

- $V_{DD} = 1.0\,\text{V}$ and $C_L = 40\,\text{fF}$.
- Each ON NMOS has resistance $R_n = 8\,\text{k}\Omega$.
- Three candidate PMOS pull-up resistances are $R_p = 40,\ 64,\ 80\,\text{k}\Omega$.
- An ON transistor is modeled as a constant resistance; an OFF transistor is an open circuit. Ignore leakage, body effect, and parasitic capacitances other than C_L. Input transitions are instantaneous.

The design must have $V_{OL} \leq 0.20\,\text{V}$ for *every* input combination that should produce logic 0.

(a) **DC behavior and sizing decisions.** **[10 points]**

Draw the transistor-level pseudo-NMOS circuit. For each candidate R_p, determine the steady-state output-low voltage for these input combinations:

$$(A, B, C) = (1, 1, 0), \qquad (0, 1, 1), \qquad (1, 1, 1).$$

Identify which pull-up resistances satisfy the V_{OL} requirement in all three cases. For each qualifying resistance, determine the static power when both pull-down paths conduct. Explain why the combination with the largest V_{OL} need not dissipate the largest static power.

(b) **Transient behavior and design recommendation.** **[10 points]**

Set $A = 1$ and $C = 0$. Let B change from 0 to 1 and, after the circuit has settled, from 1 back to 0. For the *two* pull-up resistances satisfying part (a), estimate the falling and rising propagation delays measured at $V_{DD}/2$. Account for the nonzero final low voltage.

Compare delay and static power for the two candidates. Which would you select for a timing-critical application, and which for a power-constrained application? Explain the tradeoffs.

Question 2: DCVSL Logic and Input-Transition Hazards [20 points]

Differential Cascode Voltage Switch Logic (DCVSL) uses complementary NMOS pull-down networks and cross-coupled PMOS transistors to produce complementary outputs. Design a DCVSL implementation of

$$F = C(A + B).$$

Assume $V_{DD} = 1.0\,\mathrm{V}$ and that both true and complemented forms of A, B, and C are available.

(a) **Transistor-level implementation.** [10 points]

Draw the cross-coupled PMOS devices and both NMOS pull-down networks so that the circuit produces F and $\overline{F}$. Label every gate, output, and supply connection.

For $(A, B, C) = (1, 0, 1)$ and $(0, 0, 1)$, identify the conducting pull-down network and explain how the correct complementary output values are established. Explain why, for valid complementary input signals and ideal devices, both outputs cannot remain at logic 1 in a stable state.

(b) **Transition ordering and robustness.** **[10 points]**

Consider a change from $(A, B, C) = (1, 0, 1)$ to $(0, 1, 1)$, with $C = 1$ throughout. Analyze the following two sequences, assuming sufficient time between changes for the circuit to settle:

- **Scenario I:** $A : 1 \to 0$ occurs before $B : 0 \to 1$.
- **Scenario II:** $B : 0 \to 1$ occurs before $A : 1 \to 0$.

For each scenario, find the intermediate and final values of F and $\overline{F}$ and sketch qualitative output waveforms with propagation delays indicated. Identify any unnecessary output transition and explain its origin in terms of input ordering, pull-down connectivity, and cross-coupled feedback.

Propose one circuit-level or logic-level method to prevent the unwanted transition and discuss its timing, area, or power cost.

Question 3: Edge-Triggered Registers and Timing Constraints [20 points]

Consider a positive-edge-triggered register made from two level-sensitive latches. The master is transparent while $CLK = 0$, and the slave is transparent while $CLK = 1$. Assume complementary, nonoverlapping clock phases.

(a) **Master-slave operation and waveform analysis.** **[10 points]**

At $t = 0$, $CLK = 0$, $Q_M = 0$, and $Q = 0$. The clock rises at 100, 300, and 500 ps and falls at 200, 400, and 600 ps. Input D is initially 0 and changes as follows:

Time (ps)	New value of D
50	1
145	0
220	1
270	0
335	1
455	0
475	1

For this part only, assume ideal latches, zero propagation delay, and no setup or hold violations. Draw CLK, D, Q_M, and Q from 0 to 550 ps. Mark the transparency intervals, state Q immediately after every rising clock edge, and explain why a change in D during the high clock phase does not immediately affect Q.

(b) **Setup/hold timing and corrective design.** **[10 points]**

Two positive-edge-triggered registers R_1 and R_2 are connected by a combinational path. R_1 launches data and R_2 captures it on the next rising edge. The nominal clock period is 500 ps, with the following parameters:

Parameter	Value
Maximum clock-to-Q delay of R_1	75 ps
Minimum clock-to-Q delay of R_1	30 ps
Maximum combinational delay	385 ps
Minimum combinational delay	25 ps
Setup time of R_2	65 ps
Hold time of R_2	45 ps
Clock skew	+20 ps

Positive skew means the clock reaches R_2 20 ps after R_1; assume no clock uncertainty and the same skew on consecutive clock edges. Determine setup and hold slacks, and identify violations.

A delay buffer adds 20 ps to *both* the minimum and maximum combinational path delays. Determine the new setup and hold slacks, whether the hold problem is fixed, and the minimum permissible clock period after inserting the buffer. Explain why increasing the clock period alone does not correct a hold violation.

Question 4: Cadence Virtuoso: NAND3/NOR3 Sizing and Transistor Fingering [40 points]

Use **Cadence Virtuoso** and the **FreePDK45** process to design schematic-only versions of a three-input NAND (NAND3) and a three-input NOR (NOR3). No physical layout is required for this homework.

Use $V_{DD} = 1.0\,\text{V}$ and $C_L = 32\,\text{fF}$. Minimum transistor length and width (for either NMOS or PMOS) are $L_{\min} = 50\,\text{nm}$ and $W_{\min} = 90\,\text{nm}$. Do not choose devices below these limits.

For each gate under test, drive one input from a separate inverter. Drive that inverter with a square-wave source with period 5 ns, rise and fall times 50 ps, and 50% duty cycle. Hold the other two inputs at noncontrolling values so that the output toggles: tie them high for NAND3 and low for NOR3. Measure delays between the 50%-V_{DD} crossings at the *input of the gate under test* and at its output.

(a) **Schematic design, sizing, and delay matching.** [20 points]

Construct the NAND3 and NOR3 transistor-level schematics in FreePDK45. Size the NMOS and PMOS devices to make τ_{HL} and τ_{LH} approximately equal for each gate. Your target is also to match each gate's delay to the *fanout-of-four (FO4) delay of a reference inverter*. Characterize the FO4 inverter using the same process, supply, transition definitions, and comparable test conditions; explain how you constructed the FO4 load.

Choose and report the transistor dimensions of the reference inverter and input-driver inverters, and keep them fixed throughout the comparisons. Construct the FO4 load by connecting the inputs of four identical copies of the reference inverter to its output. The 32 fF load applies only to the NAND3 and NOR3 outputs; do not add it to the FO4 test.

Report the final transistor dimensions, τ_{HL}, τ_{LH}, average gate delay, and FO4 reference delay for each gate. Quantify any remaining mismatch rather than assuming exact equality is achievable.

(b) **Transistor fingering experiment and documented results.** [20 points]

Use transistor fingering for *both* gates. Sweep at least three valid numbers of fingers for each design. For a controlled comparison, keep the total effective transistor width fixed while varying the number of fingers wherever the FreePDK45 device model permits. If the model imposes a different convention, explain how width and finger count were specified.

For both NAND3 and NOR3, plot *gate delay* (vertical axis, in ps) against *number of fingers* (horizontal axis). Show the average delay and, if possible, the individual τ_{HL} and τ_{LH} values. Explain whether delay improves, worsens, or remains unchanged with finger count, and discuss what schematic-level simulation can and cannot capture about physical transistor fingering.

Include all of the following in your submission:

- Clearly labeled, white-background screenshots of both schematics, showing device dimensions and finger counts.
- Transient waveforms for both designs: V_{in} versus time and V_{out} versus time, with the measured delay crossings identified.
- A table listing τ_{HL}, τ_{LH}, and average delay for each design and each finger count, as well as the FO4 reference comparison.
- Two delay-versus-finger-count graphs, one for NAND3 and one for NOR3, together with a short interpretation of the results.